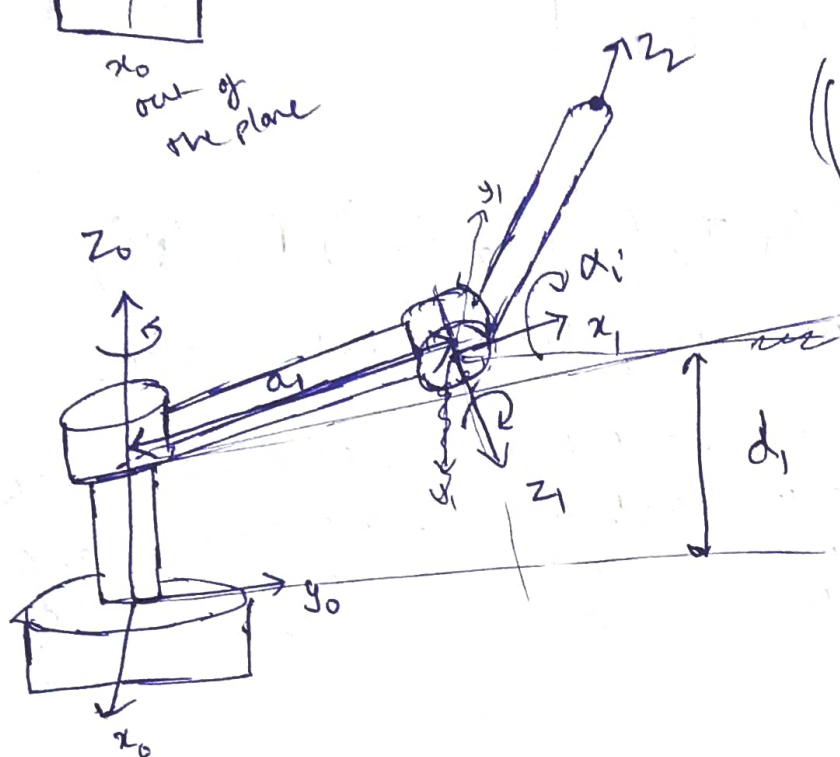
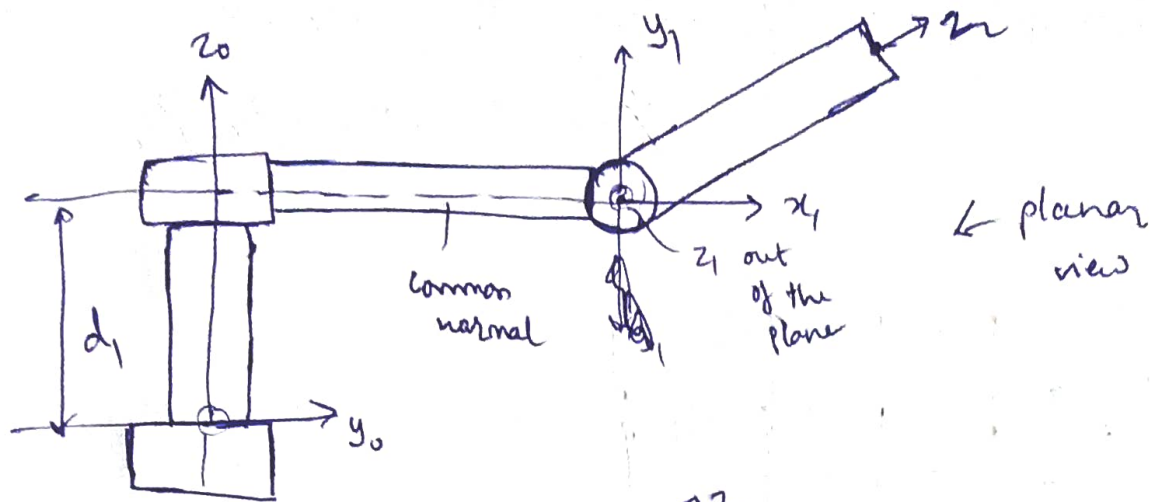


2 DOF non-planar manipulator FK



$(\alpha_i = -90^\circ \text{ deg in this case})$

$\theta_i = \text{rotation about } z_{i-1} \text{ viz } z_0$

$= 90^\circ \text{ deg in this case}$
(a variable)

But for generality consider

$a_1, \alpha_1, d_1, \theta_1$

$\alpha_1 = \text{common normal between } z_1 \text{ \& } z_2$

Now for Transformation T_1^0

The four parameters are $\theta_1, d_1, a_1, \alpha_1$

$-a_1 = \text{distance from } z_0 \text{ to } z_1 \text{ along } x_1$

$\alpha_1 = \text{angle between } z_0 \text{ \& } z_1 \text{ about } x_1$

$d_1 = \text{displacement from } O_0 \text{ (origin) till the common normal level along } z_0$

$\theta_1 = \text{angle between } x_0 \text{ and } x_1 \text{ about } z_0$
(measured $0 \rightarrow 1$)

same way end effector frame is {2}

and transformation from {1} to {2} involves

$$a_2, \alpha_2, d_2, \theta_2$$

DH table

i	a_i	α_i	d_i	θ_i
1	a_1	α_1	d_1	θ_1
2	a_2	α_2	d_2	θ_2

for a standard DH:

$$T_i^{i-1} = R_z(\theta_i) T_z(d_i) T_x(a_i) R_x(\alpha_i)$$

Multiplying those 4 elementary transformations gives

$$T_i^{i-1} = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now,

$$T_1^0 \rightarrow a_1, \alpha_1, d_1, \theta_1$$

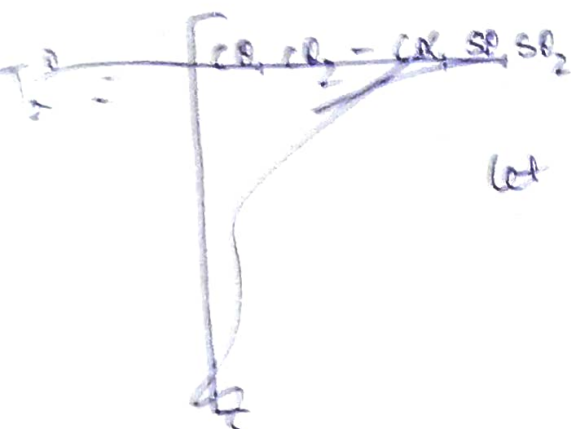
$$\therefore T_1^0 = \begin{bmatrix} c\theta_1 & -s\theta_1 c\alpha_1 & s\theta_1 s\alpha_1 & a_1 c\theta_1 \\ s\theta_1 & c\theta_1 c\alpha_1 & -c\theta_1 s\alpha_1 & a_1 s\theta_1 \\ 0 & s\alpha_1 & c\alpha_1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^1 \rightarrow a_2, \alpha_2, d_2, \theta_2$$

$$\therefore T_2^1 = \begin{bmatrix} c\theta_2 & -s\theta_2 c\alpha_2 & s\theta_2 s\alpha_2 & a_2 c\theta_2 \\ s\theta_2 & c\theta_2 c\alpha_2 & -c\theta_2 s\alpha_2 & a_2 s\theta_2 \\ 0 & s\alpha_2 & c\alpha_2 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now, $T_2^0 = T_1^0 T_2^1$

we get



let $C_1 = \cos \theta_1$ same for $\sin$
 $C\alpha_1 = \cos \alpha_1$

$$T_2^0 = \begin{bmatrix} C_2 - C\alpha_1 S_1 S_2 & -C_1 C\alpha_2 S_2 - C\alpha_1 C\alpha_2 S_1 C_2 & C_1 S\alpha_2 S_2 + C\alpha_1 S_1 C_2 S\alpha_2 & a_1 C_1 + a_2 C_1 C_2 - a_2 C\alpha_1 S_1 S_2 + d_2 S\alpha_1 S_1 \\ S_1 C_2 + C_1 C\alpha_1 S_2 & -C_1 C\alpha_1 C_2 C\alpha_2 - C_1 S\alpha_1 S\alpha_2 - S_1 S_2 C\alpha_2 & -C_1 C\alpha_1 C_2 S\alpha_2 - C_1 S\alpha_1 C\alpha_2 + S_1 S_2 S\alpha_2 & a_1 S_1 + a_2 C_1 C\alpha_2 + a_2 S_1 C_2 - C_1 d_2 S\alpha_1 \\ S\alpha_1 S_2 & S\alpha_1 C_2 C\alpha_2 + C\alpha_1 S\alpha_2 & -S\alpha_1 C_2 S\alpha_2 + C\alpha_1 C\alpha_2 & d_1 + a_2 S\alpha_1 S_2 + d_2 C\alpha_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

for our particular case

$\alpha_1 = -90^\circ$ $\alpha_2 = 0^\circ$

θ_1, θ_2 joint variables

$$T_2^0 = \begin{bmatrix} C_1 C_2 & -C_1 S_2 & -S_1 & a_1 C_1 + a_2 C_1 C_2 - d_2 S_1 \\ S_1 C_2 & -S_1 S_2 & C_1 & a_1 S_1 + a_2 S_1 C_2 + d_2 C_1 \\ -S_2 & -C_2 & 0 & d_1 - a_2 S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$